

Magnetization Evolution in Network Models of Porous Rock under Conditions of Drainage and Imbibition

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Surface-enhanced relaxation of nuclear magnetization in fully and partially saturated water-wet porous media is studied using a pore network simulator. The simulator is based on a description of the pore space in terms of a regular cubic lattice of pores and throats following respective size distributions. The latter are obtained from geometric characterization of 3D stochastic replicas of real porous rock samples. Concepts of percolation theory are used to simulate the pore-scale distribution of a strongly wetting phase under conditions of quasistatic drainage by or imbibition against a nonwetting phase. The results of these simulations compare favorably with experimental mercury intrusion/retraction curves. The equations governing magnetization evolution in a connected pore system are then solved by matrix diagonalization for different values of wetting-phase saturation along the primary drainage and secondary imbibition paths. Analysis of the corresponding spectra of decay rates provides new insights regarding the influence of pore structure (pore and throat size distributions, spatial correlation), surface relaxation strength, and fluid distribution on diffusive coupling between pores. For pore and throat size distributions and surface relaxation strength representative of sandstones, diffusive coupling is quite important, especially under conditions of partial saturation. Remarkably, simulations show that correlated heterogeneity is the main reason pores appear poorly coupled with respect to NMR relaxation—an assumption underlying the correspondence between pore size and relaxation time distributions. Finally, for a given value of wetting-phase saturation, the history of saturation change (drainage or imbibition) is shown to have a profound effect on the spectrum of decay rates. © 2002 Elsevier Science (USA)

Key Words: magnetic resonance; relaxation; spatial correlation; diffusion; drainage; imbibition; network model; pore size distribution.

1. INTRODUCTION

The sensitivity of nuclear magnetic resonance (NMR) to molecular-level events within fluids provides important opportunities for determining fluid states and properties within porous media (1). NMR methods have recently found important practical applications in downhole well logging for hydrocarbon formation evaluation (2). In a NMR measurement, polarized

nuclear spins in fluid molecules probe the pore space by diffusive motion. The magnetic moment carried by these spins relaxes due to dipolar interactions with other fluid molecules and, for a wetting fluid, also due to interactions at the fluid-solid interface. The total magnetization evolves by diffusion and relaxation processes and is sensitive to the geometry of the pore space (3). Measurements of the decay of nuclear magnetization are relatively straightforward. However, interpretation of these measurements to determine structural properties of porous media and characterize the distribution and flow of fluids therein is a task that remains partly unresolved (4).

In a seminal contribution, Brownstein and Tarr (5) established that the effect of surface-enhanced relaxation on magnetization decay within a single isolated pore of simple geometry is described, to an excellent approximation, by a single relaxation time, T_s , depending on the value of the dimensionless parameter $\rho\alpha/D$,

$$\frac{1}{T_s} = \begin{cases} \frac{\rho}{\alpha}, & \frac{\rho\alpha}{D} \ll 1 \\ \frac{D}{\alpha^2}, & \frac{\rho\alpha}{D} \gg 1 \end{cases}, \quad [1]$$

where D is the self-diffusion coefficient of the saturating fluid, ρ is a parameter characterizing the strength of surface relaxation, and α is a characteristic pore size corresponding to V_p/A_p , the ratio of pore volume to pore surface area. The condition $\rho\alpha/D \ll 1$ corresponds to the so-called “fast-diffusion” regime—a situation for which the relaxation rate is controlled by pore surface interactions. The condition $\rho\alpha/D \gg 1$ corresponds to the “slow-diffusion” regime—a situation for which the rate of surface relaxation equals the rate of molecular diffusion to the fluid-solid interface.

Bulk relaxation processes within the fluid contribute in parallel to surface relaxation, so that the observed relaxation rate is given by (1)

$$\frac{1}{T} = \frac{1}{T_s} + \frac{1}{T_b}, \quad [2]$$

where $1/T_b$ is the bulk relaxation rate. Bulk relaxation is often, but not always (6), negligible by comparison to surface relaxation (i.e., $T_b \gg T_s$). In any case, its effect can be easily

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separated from the effect of surface relaxation on magnetization decay and, for this reason, it is not discussed further.

The interpretation of magnetization decay observed in porous rock is based on two fundamental, and seemingly contradictory, assumptions (2). The first assumption is that each pore is in the “fast-diffusion” regime; i.e., diffusion is fast enough to maintain a uniform magnetization across space. This assumption is essential for establishing a direct relationship between relaxation time and pore size, as shown in Eq. [1]. Sufficiently large pores may be in the slow-diffusion regime, although in such cases it is likely that the longer surface relaxation will be swamped by T_b (2). For a variety of rock samples, Latour *et al.* (6) obtained relaxation time distributions that were independent of temperature over the range 25–175°C. Since the relaxation time in the diffusion-limited case depends explicitly on the bulk diffusion coefficient (cf. Eq. [1]), itself a function of temperature, the results of Latour *et al.* (6) suggest that the samples studied were in the fast-diffusion regime.

The second assumption is that diffusion between pores is unimportant, so that magnetization in each pore decays independently of other pores. On this assumption, the broad distribution of relaxation times usually observed in samples of porous rock (6–8),

$$M(t) = M_0 \sum_i f_i e^{-t/T_i}, \quad [3]$$

is interpreted as a volume-based distribution of pore sizes, where f_i is the fraction of pore volume associated with pores of characteristic size $(V_p/A_p)_i = \rho T_i$ (9). Since, however, the pore space is highly connected, it is reasonable to expect that nuclear spins will probe more than one pore during NMR observation times (3), in apparent contradiction with the assumption of weak coupling.

Following Cohen and Mendelson (10), McCall *et al.* (11) investigated the effects of diffusive coupling between pores under conditions of fast diffusion. These authors formulated and solved the coupled magnetization balance equations for a fully saturated network of pores connected with pore throats. They demonstrated that the breadth of the decay rate distribution decreases and the distribution shifts to higher rates, as coupling increases. For very strong coupling, the spectrum is reduced to a narrow peak at a rate determined by the ratio of the total surface to total volume of the pore system. This situation corresponds to a magnetization that is uniform within the entire sample and decays as a single exponential.

McCall *et al.* (11) argued that pore coupling in sandstone rocks is weak, and therefore the assumption that the spectrum of decay rates reflects the density of pore sizes is probably a good one. Their conclusions, however, were based on simulations using random pore networks in which the size of pore throats was uniform. In real sandstone rock and other porous media, not only pore and throat sizes follow respective distributions, but also correlation between the size of a throat and the size of

adjacent pores (12, 13), as well as spatial correlation among the pore sizes (14–16), are very likely to exist. None of these correlations were included in the model of McCall *et al.* (11). Their theoretical prediction that a significant narrowing of the decay rate spectrum should take place as the coupling strength is increased by increasing the bulk diffusion coefficient (i.e., by increasing temperature) was not confirmed by the experiments of Latour *et al.* (6).

Importantly, the assumption that a relaxation time distribution corresponds to a pore size distribution also underlies attempts to estimate a drainage capillary pressure curve by simply rescaling a relaxation time distribution (17). This correspondence has received little theoretical attention. NMR relaxation measurements in porous samples that have undergone drainage-type displacement have revealed qualitative changes in the spectrum of decay rates, which are consistent with the accepted view that (i) relaxation time is proportional to pore size for water-saturated pores and (ii) the nonwetting fluid enters progressively smaller pores as the water saturation decreases (4).

Network models have been instrumental in understanding the distribution and flow of immiscible fluids in porous media (18, 19). However, with the exception of McCall *et al.* (11), who only considered fully saturated networks of very simple and somewhat unrealistic geometry, NMR relaxation in porous media has not been studied by network simulation. This task is undertaken here. We are particularly interested in network models which are consistent with structural information extracted by analysis of 3D images of the pore space and which can quantitatively describe experimental mercury porosimetry data (20). With such models at hand, our attention is then focused on information about the pore geometry and fluid distribution that may be garnered by NMR relaxation analysis.

This paper is structured as follows. In the first section we describe the construction of the pore network model, paying particular attention to its geometric and correlation aspects (pore and throat size distributions, pore/throat correlation, spatial structure, pore surface roughness). In the same section we briefly describe the algorithms used to simulate drainage and imbibition, develop the magnetization balance equations, and discuss their solution. In a subsequent section, we first demonstrate that the network model can successfully simulate experimental mercury-air drainage and imbibition capillary pressure curves obtained from Berea sandstone and North Sea chalk samples. We then analyze magnetization decay spectra obtained at different values of saturation along the primary drainage and secondary imbibition paths. We show that, in the absence of spatial correlation among the pores and for pore structure and relaxation parameters consistent with experimental data for Berea sandstone, diffusive coupling is quite pronounced. We then demonstrate that spatial correlation of the pore sizes masks the effects of diffusive coupling, so that the observed spectrum of decay rates at $S_w = 1$ corresponds well with the pore size distribution, even when diffusive coupling is significant. We also discuss the significance of these results for the interpretation of NMR relaxation spectra

at partial saturation under conditions of primary drainage and secondary imbibition. Finally, we use the network simulator to explore the correspondence between NMR relaxation spectra at $S_w = 1$ and primary drainage capillary pressure curves.

2. THEORY

2.1. Network Model of Rock Pore Structure

Following previous work (21, 22), we choose to represent the structure of porous rock with a regular cubic lattice of pores and throats. Pores are identified with sites in the lattice and throats with bonds between sites. We consider that pores are irregular in shape and choose to characterize them by an equivalent-cube dimension, R_p , so that their pore volume and surface area are given by $V_p = 8R_p^3$ and $S_p = 24R_p^2$, respectively. We also assume that pore sizes are randomly distributed on lattice sites according to a known pore size distribution (PSD), in an uncorrelated or spatially correlated fashion. In the latter case, we choose an isotropic exponential autocorrelation function with correlation length equal to five lattice sites (15, 22).

Consistent with recent advances in 3D pore space characterization (23, 24), we take pore throats to represent volumeless constrictions between pores. The throat cross sections may also be irregular. We choose to characterize pore throats in terms of a single dimension, R_t , such that $R_t = \sqrt{A_t}/2$, where A_t is the throat cross-sectional area. The distributions of both A_t and V_p can be measured, if a description of the pore space in the form of 3D volume data is available (23, 24). Determination of the characteristic dimensions R_p and R_t from such information is straightforward. Throat sizes from a respective distribution (TSD) are distributed on the bonds of the lattice according to the bond-site correlation scheme detailed elsewhere (21, 25). This scheme introduces positive correlation between the size of a throat and the size of the pores communicating through it, in agreement with the generally accepted view of rock pore structure (13). To account for resistance to molecular diffusion between pores in the model, we consider an interpore diffusion length distribution (DLD). Interpore diffusion lengths correspond to the lengths of distinct pore space channels. Their determination by morphological skeletonization of 3D volume data has been detailed elsewhere (23). Interpore diffusion lengths are also positively correlated to the size of the adjoining pores. Pore size, throat size, and interpore length distributions obtained from 3D stochastic models representative of Berea sandstone are shown in Fig. 1.

2.2. Percolation Modeling of Drainage and Imbibition

Assuming strongly water-wet conditions, we model quasistatic drainage as bond-correlated site percolation without trapping (12, 15, 25), and quasistatic imbibition as mixed-percolation (22) on a lattice. The reader is referred to the original publications for detailed descriptions of the algorithms used to determine the fluid occupancy of pores and throats as a function of capillary pressure during drainage and imbibition. Here, we

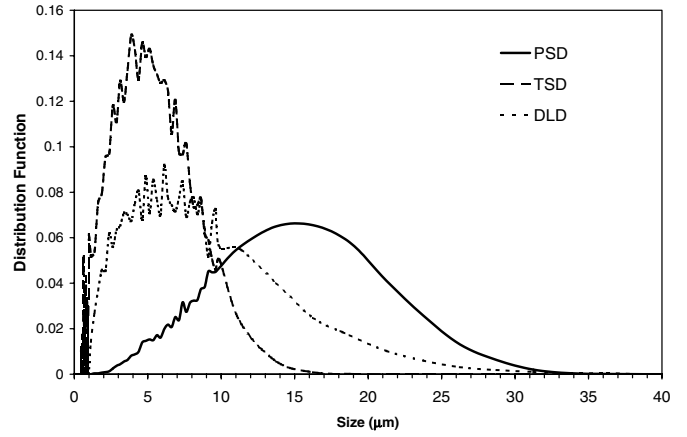


FIG. 1. Number-based pore size (PSD), throat size (TSD), and diffusion length (DLD) distributions for Berea sandstone.

note that the critical capillary pressure for piston-type invasion of the nonwetting phase into a pore or throat during drainage is computed from

$$P_c^{(piston)} = \frac{\sigma f(\theta_R)}{R}, \quad [4]$$

where θ_R is the receding contact angle, σ is the interfacial tension and the function $f(\theta_R)$ is given by (26)

$$f(\theta_R) = \frac{\theta_R + \cos^2 \theta_R - \pi/4 - \sin \theta_R \cos \theta_R}{\cos \theta_R - \sqrt{\pi/4 - \theta_R + \sin \theta_R \cos \theta_R}}. \quad [5]$$

The critical capillary pressure for piston-type retraction of the nonwetting phase from a pore or throat during imbibition is also computed from Eq. [4] and Eq. [5] using the advancing contact angle θ_A in place of θ_R . Cooperative pore filling (27, 28) is not taken into account.

The critical capillary pressure for snap-off in pore throats during imbibition is computed as

$$P_c^{(snap-off)} = \frac{P_c^{(piston)}}{\xi}, \quad [6]$$

where the critical pressure ratio, ξ , is defined as the ratio of capillary pressure for piston-type retraction to the capillary pressure for snap-off in a pore throat (29). For throats and pores of simple geometry, the critical pressure ratio ξ can be computed from interfacial stability considerations (22, 30). However, to avoid imposing arbitrary geometric restrictions and to preserve the flexibility of the network model, we choose to treat ξ as an adjustable parameter in the simulation (31). We expect that, if the model provides a reasonable representation of real media, the values of ξ which give good agreement with experimental data will also agree reasonably well with theoretical predictions of this parameter (22, 30).



FIG. 2. Illustration of irregular pore geometry in stochastically reconstructed porous media. Pore throats are shown in black (after (24)).

2.3. Modeling the Effect of Irregular Pore Geometry on Fluid Distribution

It is important to note that the rock-pore interface in real porous media is quite rough and often fractal (32–34). Accordingly, pore and throat sizes may be expected to span a broad range of length scales. In agreement with this expectation, experimental mercury porosimetry curves usually reveal that considerable fractions of the total pore volume are intruded over a very broad range of pressures. These observations have two important consequences for the simulation of drainage and imbibition using network models. First, PSD and TSD functions determined by analysis of 3D volume data of limited resolution (ca. 1–5 μm for sandstone and 0.1–0.3 μm for chalk) and subsequently used in network models do not represent the entire spectrum of pore and throat sizes present in real rock. Secondly, even if the pore volume distribution were perfectly known, mapping of irregular pores (see Fig. 2) onto regular, albeit angular, objects (e.g., cubes) would still result in poor approximation of the volume of the wetting fluid that remains in the pores after they are invaded by a nonwetting fluid. Tsakiroglou and Payatakes (35) recognized this and proposed to endow network models with a geometrically determined fractal model of the pore surface, in connection with simulations of mercury porosimetry.

Here, we propose to capture the effect of pore structure that is not explicitly represented in the network model, on the distribution of wetting fluid at the pore scale as follows. Consider that upon invasion of the nonwetting fluid into pore i at capillary pressure $P_{c,i}^*$, the remaining wetting fluid occupies a fraction S_w^* of the volume of the pore. At capillary pressure $P_c > P_{c,i}^*$, the fraction of pore volume occupied by the wetting fluid is assumed to vary according to the expression:

$$\frac{S_{w,i}}{S_w^*} = \left(\frac{P_{c,i}^*}{P_c} \right)^\eta. \quad [7]$$

The initial fraction S_w^* and “filling” exponent η are physically meaningful model parameters. For $0 < \eta < 1$, drainage of the wetting fluid within each pore at $P_c > P_{c,i}^*$ corresponds to filling with the nonwetting phase of a surface fractal with dimension $2 < D_s < 3$. This follows from consideration of a fractal distribution of length scales characterizing the geometry of the pore walls (34). For $\eta > 1$ the pore behaves as a rough, but not fractal, object. For example, $\eta = 3$ for wetting fluid in the corner of a cube and $\eta < 3$ for more irregular (but nonfractal) geometries. The parameter S_w^* may be identified with the fraction of pore volume contributed by length scales not explicitly represented in the network model, because they are lacking from the measured TSD and PSD and/or because the measured volume of each pore is assigned to a regular geometric object. The value of $P_{c,i}^*$ is different for each pore and depends on the adjoining throat sizes and pore accessibility. During primary imbibition, and for as long as piston-type retraction or disconnection of the nonwetting fluid in a pore has not taken place, the wetting fluid content of a pore increases according to Eq. [8]. This implies neglect of nonwetting phase trapping within the unresolved pore space. Note finally that, unlike the roughness model of Tsakiroglou and Payatakes (35), the model proposed here does not require explicit description of the geometry of roughness features.

2.4. Modeling of Magnetization Evolution in the Pore Network

Following McCall *et al.* (11), we assume that magnetization within each pore is uniform (fast-diffusion regime), but magnetization exchange by diffusion between pores is possible. A balance of magnetization M_i may be written for each pore i , expressing the fact that the net rate of magnetization decay in each pore is determined by magnetization relaxation in the vicinity of the fluid-matrix interface and by diffusion to and from nearest neighboring (NN) pores,

$$\frac{dM_i}{dt} = -\rho \cdot A_{p,i} \cdot \frac{M_i}{V_{w,i}} + D \cdot \sum_j^{NN} \frac{A_{w,ij}}{L_{ij}} \cdot \left(\frac{M_j}{V_{w,j}} - \frac{M_i}{V_{w,i}} \right), \quad [8]$$

where $A_{p,i}$, $V_{w,i}$ are the surface area and volume occupied by water within pore i , respectively, D is the self-diffusion coefficient of water, and $A_{w,ij}$, L_{ij} are the cross-sectional area and length associated with the diffusion of water between pores i and j . Note that, since pore surfaces are covered with wetting fluid at all times, $A_{p,i}$ is a constant for each pore, whereas $V_{w,i}$ and $A_{w,ij}$ depend on fluid occupancy and capillary pressure. Specifically, $V_{w,i} = S_{w,i} V_{p,i}$ and $A_{w,ij} = S_{w,ij} A_{t,ij}$, where $S_{w,i}$ and $S_{w,ij}$ for pores and throats are both computed using Eq. [7].

In compact form, Eq. [8] may be written as

$$\frac{dM_i}{dt} = -\omega_\rho \cdot r_i \cdot M_i + \omega_c \cdot \sum_j^{NN} \frac{s_{ij}}{l_{ij}} \cdot \left(\frac{M_j}{v_j} - \frac{M_i}{v_i} \right), \quad [9]$$

where

$$r_i = \frac{A_{p,i}}{V_{w,i}} \cdot \frac{\langle V_w \rangle}{\langle A_p \rangle}, \quad v_i \equiv \frac{V_{w,i}}{\langle V_w \rangle}, \quad s_{ij} \equiv \frac{A_{w,ij}}{\langle A_w \rangle}, \quad \text{and} \quad l_{ij} \equiv \frac{L_{ij}}{\langle L \rangle}$$

(angular brackets denote average values). The parameters

$$\omega_\rho \equiv \rho \cdot \frac{\langle A_p \rangle}{\langle V_w \rangle} = \frac{1}{\tau_\rho} \quad \text{and} \quad \omega_c \equiv \frac{D \langle A_w \rangle}{\langle V_w \rangle \langle L \rangle}$$

represent characteristic rates of relaxation at the fluid-solid interface and of molecular diffusion between pores, respectively. Application of Eq. [9] to each pore yields an initial-value, linear ordinary differential equation problem, which is most efficiently solved by matrix diagonalization methods (11). To simplify the solution, the following change of variables is made:

$$\theta_i = \frac{M_i}{\sqrt{v_i}}. \quad [10]$$

This change in variables modifies Eq. [10] as

$$\frac{d\theta_i}{dt} = -\omega_\rho \cdot r_i \cdot \theta_i + \omega_c \cdot \sum_j^{NN} \frac{s_{ij}}{l_{ij}} \cdot \left(\frac{\theta_j}{\sqrt{v_j \cdot v_i}} - \frac{\theta_i}{v_i} \right), \quad [11]$$

or in matrix form,

$$\frac{d}{dt} \{\theta\} = -[H] \cdot \{\theta\}, \quad [12]$$

where $[H]$ is a symmetric Hamiltonian matrix. The vector of initial conditions is given as

$$\theta(0) = \frac{1}{\sqrt{N}} \cdot \begin{pmatrix} \sqrt{v_1} \\ \sqrt{v_2} \\ \vdots \\ \sqrt{v_N} \end{pmatrix}, \quad [13]$$

where N is the total number of pores in the network. The solution of the problem defined by Eq. [12] and Eq. [13] is expressed in terms of the eigenvalues and eigenvectors of matrix $[H]$ as

$$\frac{M(t)}{M(0)} = \sum_n |\theta(0) \cdot \psi_n|^2 \cdot e^{-\lambda_n t}, \quad [14]$$

$$n(\lambda) = \sum_n |\theta(0) \cdot \psi_n|^2 \cdot \delta(\lambda - \lambda_n), \quad [15]$$

where ψ_n are the eigenvectors and λ_n are the corresponding eigenvalues. Equation [15] gives the volume-weighted distribution $n(\lambda)$ of decay rates λ . In the absence of coupling between pores ($\omega_c = 0$), one has $\lambda_i = \rho A_{p,i} / V_{w,i}$ and the spectrum $n(\lambda)$ at $S_w = 1$ exactly corresponds to the volume-weighted distribution of inverse pore size.

3. RESULTS AND DISCUSSION

Determining all the eigenvalues and eigenvectors of large sparse matrices is a numerically intensive task. Due to computational constraints, we limited our investigation to pore networks of size 15^3 in 3D (without spatial correlation of pore sizes) and 50^2 in 2D (with and without spatial correlation of pore sizes). To obtain statistically representative results, we studied 10 realizations of each pore network and averaged the results. Key properties of the pore networks used are listed in Table 1.

3.1. Simulation of Drainage and Imbibition Capillary Pressure Curves

Figure 3a illustrates the effect of parameter S_w^* on the simulated drainage and imbibition capillary pressure curves for a value of $\eta = 1.5$ (networks 3D-SS1U and 3D-SS2U). The results from all 10 different realizations are plotted to illustrate the magnitude of variability among different realizations. In all simulations reported here, $\theta_R = 40^\circ$ and $\theta_A = 60^\circ$. It is observed that, as S_w^* increases, the increase of nonwetting phase saturation with capillary pressure (drainage) becomes more gradual. During imbibition, a greater amount of nonwetting fluid is displaced before a large number of piston-type and snap-off events take place. The location of the main leg of the imbibition curve is not affected, but the residual nonwetting-phase saturation is considerably reduced as S_w^* increases.

The effect of parameter η is illustrated in Fig. 3b (networks 3D-SS2U and 3D-SS3U). As can be seen, changing η from 1.5 (rough, but not fractal pore geometry) to 0.5 (fractal pore geometry) results in even more gradual drainage and imbibition curves. Reducing η has negligible effect on the residual nonwetting-phase saturation. In Fig. 3c, we compare the results from network 3D-SS3U (average over 10 realizations) to experimental mercury porosimetry curves of a Berea sandstone sample (36). Note that both drainage and imbibition curves are well reproduced by the model using $\xi = 6$, $\eta = 0.5$, $S_w^* = 0.24$, and the pore and throat size distributions shown in Fig. 1. No further attempt was made to optimize these parameters.

To further verify the simulator's ability to describe quasi-static drainage and imbibition in reservoir rock, we modeled the mercury porosimetry curves obtained experimentally from a North Sea chalk sample (37). Backscatter SEM images of this sample were used to generate a 3D stochastic replica of its

TABLE 1
Network Simulation Parameters

Model	ρ ($\mu\text{m/s}$)	ξ	S_w^*	η	PSD, TSD	Spatial correlation
3D-SS1U	—	6	0.24	0.5	Fig. 1	No
3D-SS2U	—	6	0.24	0.5	Fig. 1	No
3D-SS3U	5,100	6	0.24	0.5	Fig. 1	No
3D-CH1U	—	2	0.30	1.1	Fig. 4a	No
2D-SS3U	5	6	0.24	0.5	Fig. 1	No
2D-SS3C	5	6	0.24	0.5	Fig. 1	Exponential

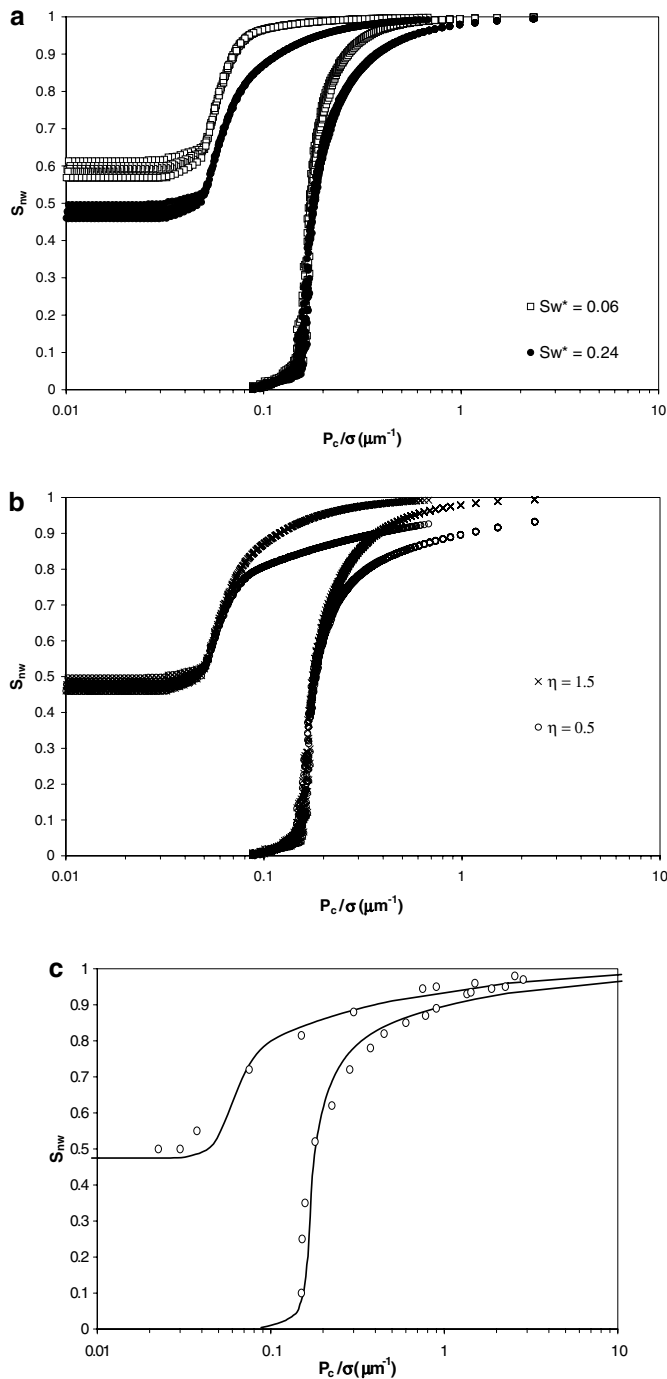


FIG. 3. (a) Effect of parameter S_w^* on simulated drainage and imbibition capillary pressure curves. (b) Effect of parameter η on drainage and imbibition capillary pressure curves. (c) Comparison of simulated and experimental mercury porosimetry curves for Berea sandstone.

microstructure, which was then analyzed using pore space partitioning techniques developed recently (23). The pore and throat size distributions measured in the 3D stochastic replica of the sample are shown in Fig. 4a. For details on the 3D reconstruction and characterization of the chalk microstructure, the reader

is referred to Talukdar *et al.* (37). As shown in Fig. 4b, network simulation of drainage and imbibition using $\xi = 2$, $\eta = 1.1$, and $S_w^* = 0.3$ (network 3D-CH1U) is in very good agreement with the experiment. Again, no rigorous optimization of these parameters was attempted.

It is encouraging that the values of critical pressure ratio ξ that give reasonable agreement with experiment are also in reasonable agreement with theoretical analyses of interfacial stability in pore throats of geometry representative of the media under consideration. A value of $\xi = 6$ for sandstone suggests the presence of short pore throats of anticlastic shape, as might be expected for consolidated packing of grains, whereas a value of $\xi = 2$ for chalk suggests the presence of elongated pore throats. Additionally, the value $\eta = 0.5$ for sandstone corresponds to a surface fractal dimension $D_s = 2.5$, in reasonable agreement with published values (38), whereas pore surfaces in chalk appear to be rough, but not fractal.

It should be noted that the aforementioned results were obtained from pore networks in which throat sizes are correlated to the sizes of adjoining pores, but pores of similar size are not

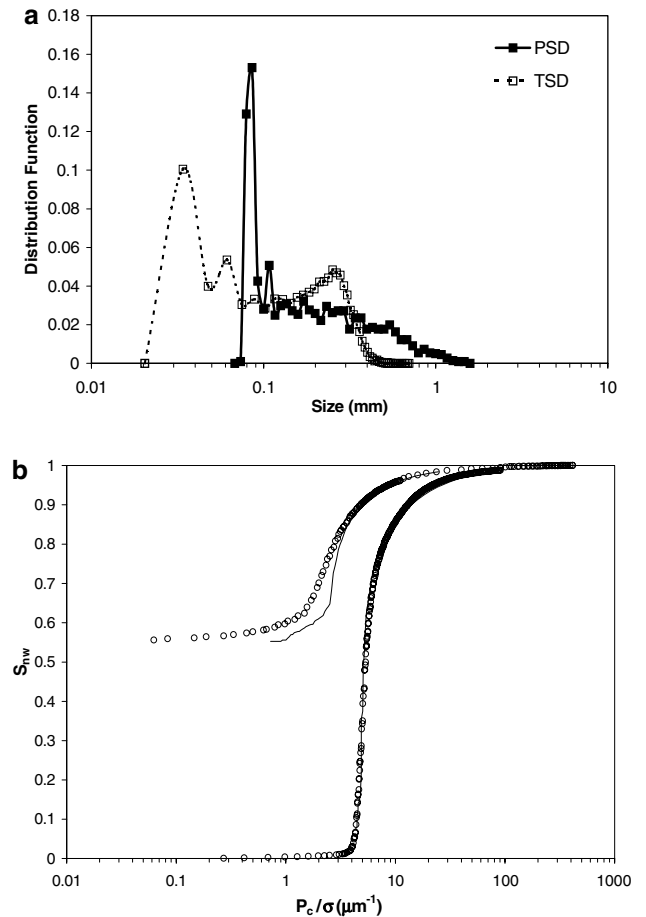


FIG. 4. (a) Number-based pore (PSD) and throat (TSD) size distributions for North Sea chalk (after (37)). (b) Comparison of simulated and experimental mercury porosimetry curves for North Sea chalk.

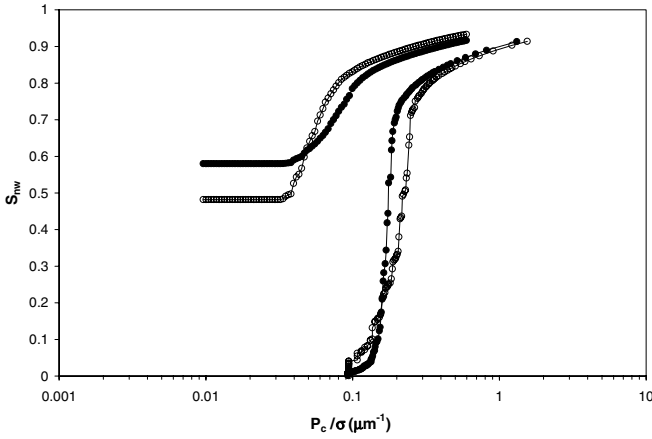


FIG. 5. Effect of spatial correlation on simulated drainage and imbibition capillary pressure curves of 2D pore networks. Solid circles, uncorrelated pore sizes; empty circles, exponentially correlated pore sizes (correlation length = 5 pores).

clustered together (no spatial correlations). In previous studies (14, 15, 22), it was found that spatial correlation among the pores causes the drainage curve to become more gradual, increases the hysteresis between drainage and imbibition and reduces the residual nonwetting-phase saturation. These observations are essentially confirmed by the results shown in Fig. 5 for networks 2D-SS3U and 2D-SS3C. Using spatially correlated networks in the modeling of experimental drainage and imbibition capillary pressure curves would likely result in a modest decrease in the values of S_w^* , η and ξ that fit the experimental data.

3.2. Decay Rate Spectra of Fully Saturated Model Pore Networks

Having established a pore network model capable of describing the drainage and imbibition capillary pressure curves of a real sample, we examined NMR relaxation at $S_w = 1$. Using the parameters corresponding to network 3D-SS3U (see Table 1), we carried out simulations using $\rho = 100 \mu\text{m/s}$ (high surface relaxivity) and $\rho = 5 \mu\text{m/s}$ (low surface relaxivity). These values are roughly at the extremes of reported surface relaxivity values, with $\rho = 5 \mu\text{m/s}$ frequently attributed to sandstones (17, 39). In each case, we compared the true decay rate spectrum to the one obtained if diffusion between the pores is ignored (i.e., $\omega_c = 0$). These comparisons are shown in Fig. 6a ($\rho = 100 \mu\text{m/s}$) and Fig. 6b ($\rho = 5 \mu\text{m/s}$). For $\rho = 100$ and $5 \mu\text{m/s}$, the mean relaxation times are $\tau_o = 0.0684$ and 1.37 s, respectively. The ratio of the two is precisely equal to the ratio of surface relaxivities, as expected. For Berea sandstone samples, Latour *et al.* (6) have reported mean relaxation times in the range 0.2–0.5 s for temperatures in the range 25–175°C. Examination of Fig. 6 reveals that, for a high value of ρ , the effects of pore coupling on the spectrum of decay rates appear to be small, whereas for a low value of ρ , the spectrum is reduced to a sharp peak. The latter result is in qualitative accord with the predictions of McCall *et al.* (11), for situations corresponding to $\omega_c \gg \omega_\rho$. If $\rho = 5 \mu\text{m/s}$ is

representative of sandstones, however, our computations show that magnetization decay should be dominated by diffusive coupling, something which has not been observed experimentally (6).

Remarkably, spatial correlations mask the effects of pore coupling on the decay rate spectra. This is born out by comparing the magnetization decay in 2D pore networks without and with spatial correlations (networks 2D-SS3U and 2D-SS3C, respectively) for a value of $\rho = 5 \mu\text{m/s}$. These results are shown in Fig. 7a for 2D-SS3U and Fig. 7b for 2D-SS3C. In the absence of spatial correlation the results are similar to those obtained from 3D networks (see Fig. 6a), as strong diffusive coupling significantly reduces the breadth of the spectrum. The effect is somewhat weaker in 2D because of reduced pore connectivity, as also noted by McCall *et al.* (11). On the contrary, when pores of similar size are clustered together, exchange of magnetization by diffusion between pores cannot effectively homogenize the kinetics of surface-enhanced relaxation (see Fig. 7b). Note the presence of a number of small peaks in the spectrum shown in Fig. 7b. Coupling is not weak but localized, and therefore its effects are not observable in the presence of spatial structure

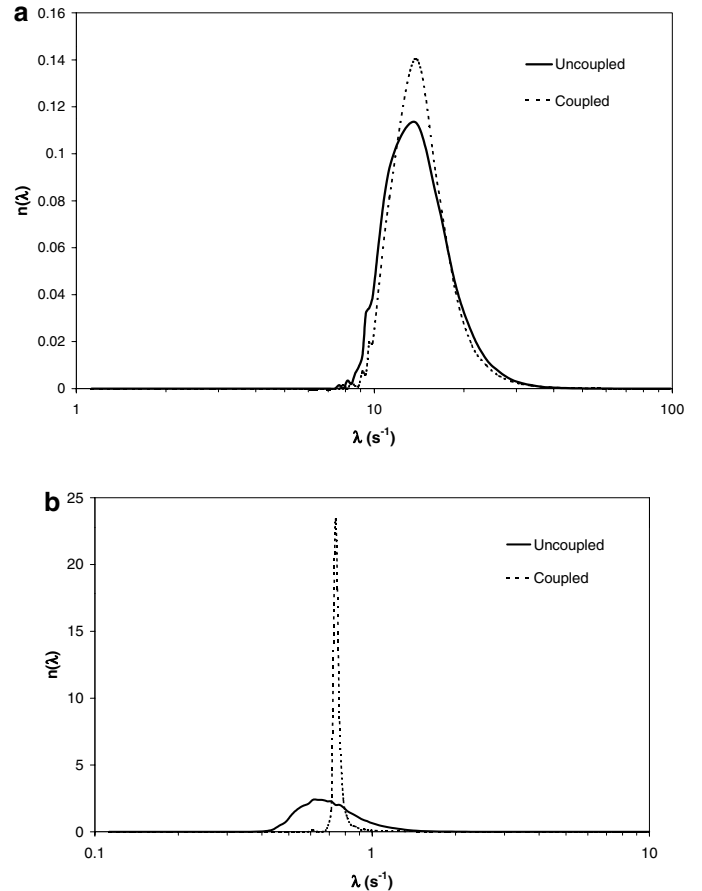


FIG. 6. (a) Magnetization decay spectra for network 3D-SS3U with $\rho = 100 \mu\text{m/s}$ ($\omega_\rho = 15.59$, $\omega_c = 0.98$). (b) Magnetization decay spectra for network 3D-SS3U with $\rho = 5 \mu\text{m/s}$ ($\omega_\rho = 0.78$, $\omega_c = 1.03$).

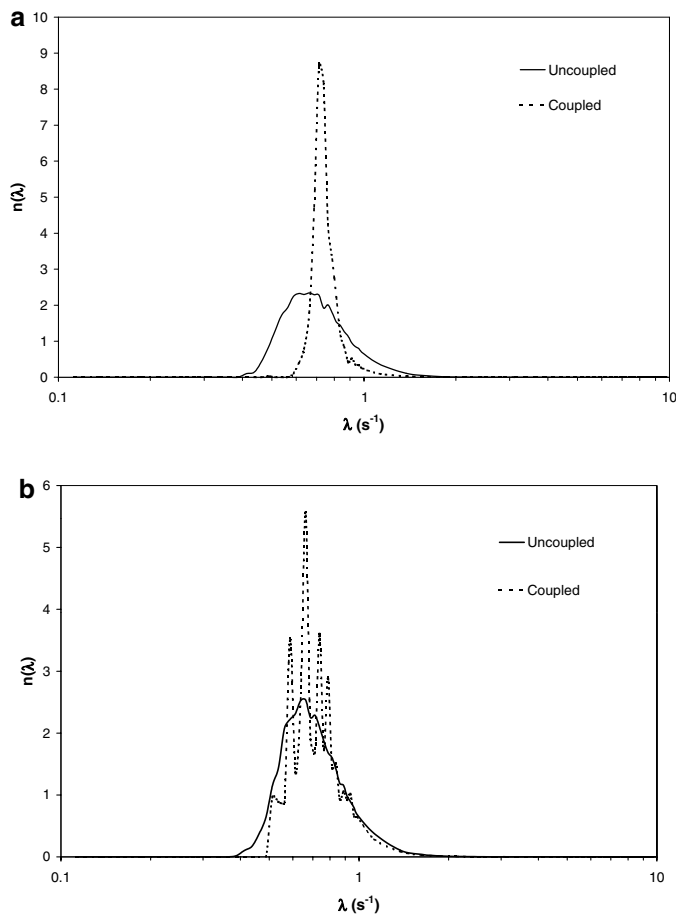


FIG. 7. (a) Magnetization decay spectra for network 2D-SS3U with $\rho = 5 \mu\text{m/s}$ ($\omega_\rho = 0.78$, $\omega_c = 0.98$). (b) Magnetization decay spectra for network 2D-SS3C with $\rho = 5 \mu\text{m/s}$ ($\omega_\rho = 0.78$, $\omega_c = 0.98$).

(i.e., locally homogeneous pore size). On the basis of these observations, we suggest that the broad relaxation rate spectra, obtained from water-saturated reservoir rock samples in numerous experimental studies, constitute evidence, not only of broad pore size distributions but also of correlated heterogeneity at length scales greater than the length scales of diffusion during NMR observation times.

As a further test, we considered the recovery of the known volume-weighted distribution of pore sizes using multiexponential inversion of the computed magnetization data, $M(t)$ (see Eq. [3]). This inversion was accomplished using a simulated annealing algorithm described in the appendix. As shown in Fig. 8, the pore size distribution obtained by multiexponential analysis of the $M(t)$ data agrees very well with the known pore size distribution. However, none of the peaks evident in the spectrum of decay rates is recovered. This is likely a consequence of the ill-conditioned nature of the multiexponential inversion problem (4). It is, thus, not surprising that such peaks are not observed in the analysis of experimental data.

3.3. Decay Rate Spectra of Partially Saturated Model Pore Networks

Network Model 3D-SS3U was used to study NMR relaxation under conditions of drainage and imbibition. This network lacks spatial structure and, on the basis of the observations made in the previous section, it is not entirely representative of Berea sandstone. Since we could not investigate NMR relaxation in large three-dimensional spatially correlated pore networks, we chose to study the problem using Model 3D-SS3U with $\rho = 100 \mu\text{m/s}$. Under such conditions, the effects of pore coupling appear to be weak (see Fig. 6a).

As seen in Fig. 9a, the larger pores are preferentially filled with nonwetting fluid during drainage. Note that we do not consider bulk relaxation effects and therefore no signal from the nonwetting phase is reported in the spectra. The wetting fluid in the invaded pores continues to be in contact with the entire surface of the pores, but its volumetric content decreases during drainage. This results in the gradual disappearance of the slower decay rates as the effective surface to volume ratio of the invaded pores increases, and the spectrum shifts toward higher rates. The remaining slow decay rates present in the spectrum belong to larger pores that are inaccessible by the nonwetting fluid.

During imbibition, the smaller pores are preferentially invaded by the wetting fluid, resulting in the gradual disappearance of the faster decay rates, as shown in Fig. 9b. The spectrum gradually shifts to slower rates as an increasingly greater amount of wetting phase exists in larger pores. As the nonwetting phase saturation decreases, the spectrum of decay rates shown in Fig. 9a for $S_w = 1$ is only partly recovered, due to the presence of residual nonwetting-phase saturation. This is more clearly shown in Fig. 10a. Note the presence of a second peak in the spectrum under conditions of residual nonwetting-phase saturation. Pores containing trapped nonwetting phase also contain

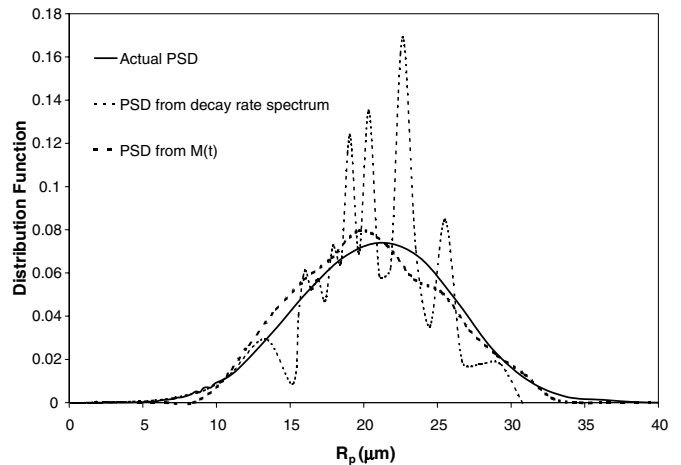


FIG. 8. Recovery of pore size distribution (PSD) from the spectrum $n(\lambda)$ and by inversion of $M(t)$ using a simulated annealing algorithm.

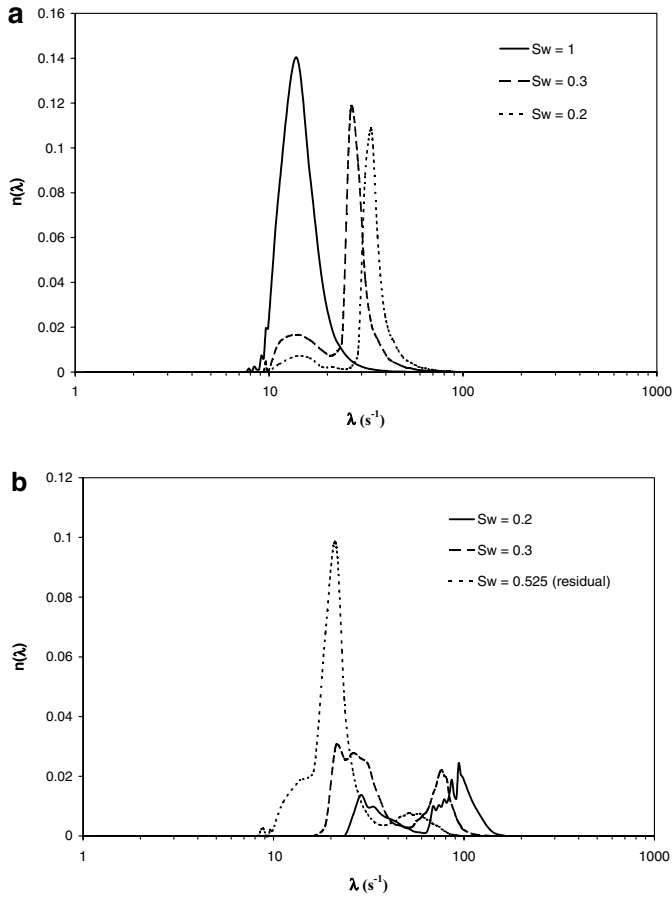


FIG. 9. (a) Magnetization decay spectra of partially saturated pore network 3D-SS3U ($\rho = 100 \mu\text{m/s}$) during drainage ($S_w = 1$, $\omega_\rho = 15.59$, $\omega_c = 0.98$; $S_w = 0.3$, $\omega_\rho = 100.49$, $\omega_c = 2.39$; $S_w = 0.2$, $\omega_\rho = 201.75$, $\omega_c = 2.92$). (b) Magnetization decay spectra of partially saturated pore network 3D-SS3U ($\rho = 100 \mu\text{m/s}$) during imbibition ($S_w = 0.2$, $\omega_\rho = 85.40$, $\omega_c = 1.30$; $S_w = 0.3$, $\omega_\rho = 54.24$, $\omega_c = 1.49$; $S_w = 0.525$, $\omega_\rho = 29.50$, $\omega_c = 1.46$).

a significant amount of wetting phase, whose characteristic relaxation rate is faster than most of the rates appearing in the spectrum at $S_w = 1$. Figure 10b compares NMR relaxation rate spectra at the same saturation ($S_w = 0.3$) for drainage and imbibition. These spectra reflect the markedly different distribution of the wetting phase in each case. During drainage, large pores not accessible to nonwetting phase contribute to the slow-rate end of the spectrum. During imbibition, many of these pores contain trapped nonwetting phase, resulting in the appearance of a second peak corresponding to fast relaxation rates in the spectrum. Clearly, interpretation of decay spectra in terms of pore structure parameters cannot be made without knowledge of the history of saturation change.

On the basis of the above observations, we investigated whether it is possible to identify the pores containing trapped nonwetting phase in the spectrum of decay rates at conditions of residual nonwetting phase saturation. A decomposition of this spectrum assuming that coupling effects are negligible is shown

in Fig. 11a. As seen in this figure, the spectrum is almost perfectly split into two portions. The slower rates are contributed almost exclusively from pores that are fully saturated with the wetting phase. If diffusion is ignored, the area under the peak corresponding to the fastest decay rates measures the fraction of pore volume occupied by wetting phase in pores containing residual nonwetting phase. Note that the slowest of these rates correspond to the largest pores containing residual nonwetting phase.

When diffusion between pores is accounted for, a very different picture emerges by decomposing the decay spectrum (see Fig. 11b). Slow decay rates are contributed by pores that are fully filled with wetting phase, as well as from pores that contain residual nonwetting phase. The same is true of fast decay rates. Large pores containing trapped nonwetting phase exchange magnetization with smaller pores fully saturated with wetting phase and their effective decay rate decreases. Similarly, large pores fully saturated with wetting phase exchange magnetization with smaller pores containing residual nonwetting phase and their effective decay rate increases. Note that these observations concern a case for which surface relaxation

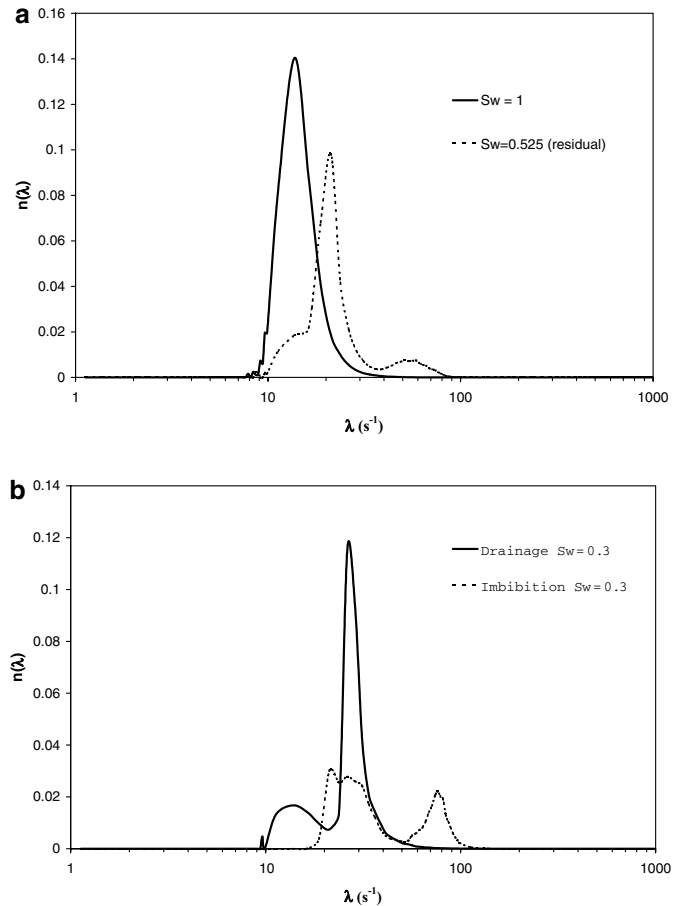


FIG. 10. (a) Comparison of magnetization decay spectra at $S_w = 1$ and $S_w = 0.525$ (residual after imbibition). (b) Comparison of magnetization decay spectra at equal saturation ($S_w = 0.3$) after drainage and imbibition.

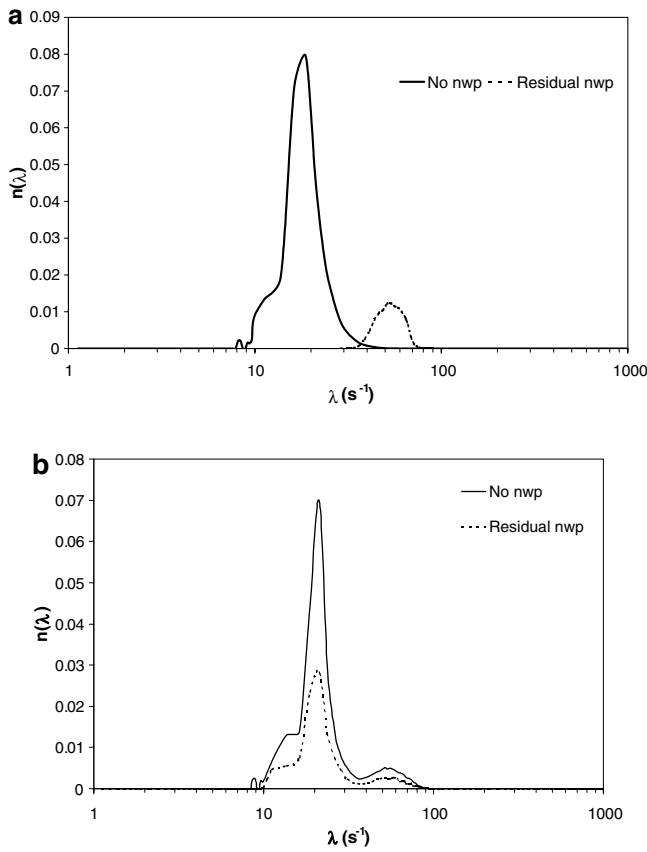


FIG. 11. (a) Decomposition of magnetization decay spectrum at S_{or} conditions in the absence of inter pore diffusion. (b) Decomposition of magnetization decay spectrum at S_{or} conditions in the presence of inter pore diffusion.

is relatively strong ($\rho = 100 \mu\text{m/s}$) and the effects of inter pore diffusion on the spectrum for $S_w = 1$ are apparently weak (see Fig. 6a). Nonetheless, diffusive coupling makes it impossible to distinguish pores containing trapped nonwetting phase in the spectrum. The same qualitative picture was obtained from simulations using Model 2D-SS3C.

3.4. Relationship between Decay Rate Spectrum at $S_w = 1$ and Primary Drainage Capillary Pressure Curve

Assuming fast-diffusion and weak effects of diffusive coupling, Kleinberg (17) suggested that relaxation rate spectra at $S_w = 1$ may be used to estimate primary drainage capillary pressure curves of porous rock. This suggestion is based on the additional assumptions that: (i) all pores of any given size are accessible by mercury at their corresponding entry pressure, and (ii) there is correlation between pore and throat sizes, so that entry pressures, which are dictated by pore throat size, are directly related to the size of adjoining pore bodies. These assumptions suggest a scaling between drainage capillary pressure and relaxation rate, such that $\kappa = P_c/\lambda$ is a constant. We tested this hypothesis using the spectra of decay rates obtained by simulation of the magnetization decay in Models 3D-SS3U ($\rho = 100 \mu\text{m/s}$,

Fig. 6a) and 2D-SS3C ($\rho = 5 \mu\text{m/s}$, Fig. 7b) at $S_w = 1$. The results are shown in Fig. 12. In each case, a good match of the simulated drainage capillary pressure curves is obtained in the approximate range $0.3 < S_w < 1$ with scaling factor $\kappa \approx 1$ (Model 3D-SS3U) and $\kappa \approx 20$ (Model 2D-SS3C). The ratio of

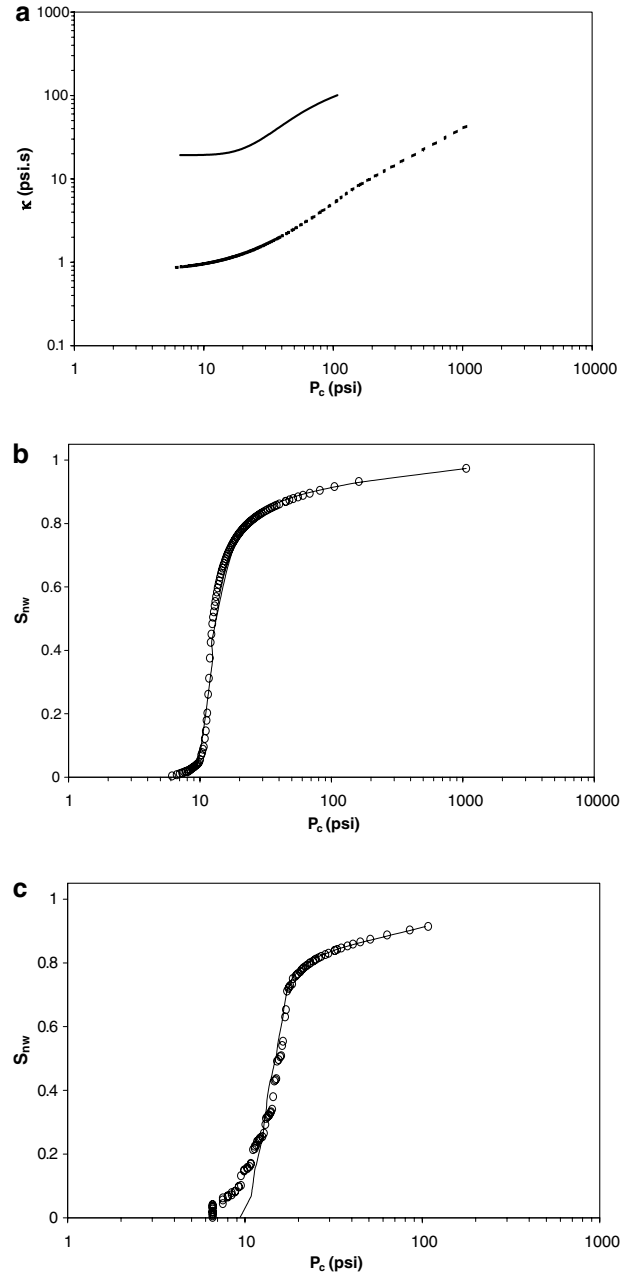


FIG. 12. (a) Scaling factor κ between relaxation rate and capillary pressure. Solid line, Model 3D-SS3U ($\rho = 100 \mu\text{m/s}$, Fig. 6a); broken line, Model 2D-SS3C ($\rho = 5 \mu\text{m/s}$, Fig. 7b). (b) Drainage capillary pressure curve (solid line) constructed by scaling the relaxation rate spectrum at $S_w = 1$ (Model 3D-SS3U, $\rho = 100 \mu\text{m/s}$) to the simulated drainage capillary pressure curve (points). (c) Drainage capillary pressure curve (solid line) constructed by scaling the relaxation rate spectrum at $S_w = 1$ (Model 2D-SS3C, $\rho = 5 \mu\text{m/s}$) to the simulated drainage capillary pressure curve (points).

the two κ values is equal to the inverse of the ratio of the corresponding surface relaxivity values, as expected. For $S_w < 0.3$, however, the scaling factor required to bring the two distributions into coincidence is observed to increase sharply with increasing capillary pressure, not unlike what is often observed experimentally (40). This result is explained by considering that pore space features that retain wetting phase at $S_w < 0.3$ during drainage (i.e., microporosity) are here assumed to be diffusively coupled to the macropores in which they are contained. Within each pore in the network, magnetization is therefore uniform and decays as a single exponential. It is precisely for this reason that fast relaxation rates attributed to microporosity are not seen in the simulated spectra at $S_w = 1$ (see Fig. 6a).

4. CONCLUSIONS

Network modeling was employed to study NMR relaxation under conditions of drainage and imbibition in porous media. A novel aspect of this work is that the model pore networks studied were developed using microstructure information obtained by comprehensive geometric and topological characterization of 3D stochastic replicas of sandstone and chalk samples. The effects of unresolved pore space geometry on fluid distribution were taken into account using a simple but physically meaningful two-parameter model. This enabled us to model quantitatively the measured mercury-air drainage and imbibition capillary pressure curves of Berea sandstone and North Sea chalk samples. Simulations of magnetization evolution in pore networks revealed that, for pore structure and surface relaxation parameters representative of sandstones, spin diffusion between pores results in significant narrowing of the decay rate spectrum, unless spatial correlation among pore sizes is taken into account. This result offers an explanation for the apparently contradictory experimental observation of broad relaxation time distributions under conditions of fast diffusion in water-saturated reservoir rock samples. We therefore suggest that such observations are evidence of correlated heterogeneity in porous rock.

Additionally, we used the simulator to explore the relaxation response of partially saturated porous rock. The results reveal the appearance in the drainage spectra of decay rates faster than any of the rates observed at $S_w = 1$. These rates were seen to correspond to the decay of magnetization in partially drained small pores. We suggest that similar observations in experimental systems may be taken as an indication of diffusive coupling of magnetization between macro- and microporosity. Furthermore, it was observed that relaxation spectra at partial saturation are very sensitive to saturation history (drainage vs imbibition). It was also shown that diffusion of magnetization between pores precludes the identification of pores containing trapped oil in the spectrum of decay rates at S_{or} . Finally, the connections between drainage capillary pressure curves and relaxation spectra at $S_w = 1$ were explored.

NOMENCLATURE

A_p	Surface area of a pore [L^2]
A_t	Cross-sectional area of a throat [L^2]
$A_{w,ij}$	Cross-sectional area occupied by wetting phase within a throat connecting pores i and j [L^2]
D	Self-diffusion coefficient of wetting phase [$L^2 t^{-1}$]
D_s	Surface fractal dimension [—]
f_i	Fraction of pore volume with relaxation time T_i [—]
L_{ij}	Length of diffusion path between pores i and j [L]
M_i	Magnetization in pore i [a.u.]
P_c	Capillary pressure [$ML^{-1} t^{-2}$]
$P_{c,i}^*$	Capillary pressure for nonwetting-phase invasion in pore i [$ML^{-1} t^{-2}$]
R_p	Equivalent pore radius [L]
R_t	Equivalent throat radius [L]
S_w	Wetting phase saturation [—]
S_w^*	Wetting phase saturation in pore upon invasion by the nonwetting phase [—]
$S_{w,ij}$	Wetting phase saturation in throat connecting pores i and j [—]
T_s	Surface relaxation time [t]
T_b	Bulk relaxation time [t]
$V_{p,i}$	Volume of pore i [L^3]
$V_{w,i}$	Volume of wetting phase within pore i [L^3]

Greek Letters

α	Characteristic pore size [L]
η	“Filling” exponent [—]
θ_A	Advancing contact angle [$^\circ$]
θ_R	Receding contact angle [$^\circ$]
λ_n	n th eigenvalue of matrix H [t^{-1}]
λ	Relaxation rate [t^{-1}]
ρ	Surface relaxivity [$L t^{-1}$]
σ	Interfacial tension [$M t^{-2}$]
ξ	Critical pressure ratio [—]
ψ_n	n th eigenvector of matrix H [a.u.]

APPENDIX

Multiexponential Inversion by Simulated Annealing

A multiexponential magnetization decay $M(t)$ may in general be written as

$$M(t) = M_o \sum_i f_i e^{-t/T_i}, \quad [\text{A.1}]$$

where f_i is the fraction of fluid volume relaxing at a rate $1/T_i$ and is constrained to be nonnegative. The established practice is to select a number of T_i values ($N = 10$ to 100), equally spaced on linear or logarithmic scales, and find the corresponding coefficients f_i (4). This problem is known to be ill-conditioned and its solution is not unique when $M(t)$ is noisy or when the mathematical computation has finite precision. To improve the

reproducibility of the results, modern inversion algorithms make use of regularization, a technique that effects a certain degree of smoothing on the resulting distributions (4).

In this work, inversion of simulated magnetization data, $M_{sim}(t)/M_o$, to obtain f_i was achieved using a simple algorithm. First, a number of T_i values ($N=26$), equally spaced in the range $[0.3125, 3.4375]$, were assigned to the midpoints of an equal number of bins. The amplitude of these bins was initialized to $f_i^{(0)}=1/N$, corresponding to an initially uniform distribution of T_i values. Simulated annealing was then implemented to minimize the objective function:

$$E^{(k)} = \sum_j \left(M_{sim}(t_j)/M_o - \sum_i f_i^{(k)} e^{-t_j/T_i} \right)^2. \quad [A.2]$$

At each iteration step k , two bins m and n ($m \neq n$, $f_m^{(k-1)} \geq \delta f$, $f_n^{(k-1)} \geq \delta f$) were randomly chosen, and their amplitudes modified as

$$f_m^{(k)} = f_m^{(k-1)} - \delta f, \quad [A.3]$$

$$f_n^{(k)} = f_n^{(k-1)} + \delta f, \quad [A.4]$$

where $\delta f \ll 1/N$. This modification altered the objective function by an amount $\Delta E^{(k)}$ given by

$$\Delta E^{(k)} = E^{(k)} - E^{(k-1)}. \quad [A.5]$$

At each iteration step k , the modification was accepted with probability $p_a^{(k)}$ given by the classical Metropolis rule,

$$p_a^{(k)} = \begin{cases} 1, & \Delta E^{(k)} < 0 \\ e^{-\Delta E^{(k)}/\psi^{(k)}}, & \Delta E^{(k)} \geq 0 \end{cases}, \quad [A.6]$$

where $\psi^{(k)}$ is a control parameter whose value decreases slowly with the number of iterations. Details on determining the initial value and rate of decrease of $\psi^{(k)}$ are given elsewhere (41). The simulated algorithm terminated when $E^{(k)}$ became smaller than a prescribed error tolerance or when no further reduction of $E^{(k)}$ could be achieved after a set number of iteration steps. Note that the conditions $f_m^{(k-1)} \geq \delta f$ and $f_n^{(k-1)} \geq \delta f$ ensured the nonnegativity of f_i . Also, $\sum_i f_i^{(k)} = 1$ at each iteration step k .

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